

RAG Chatbot Sample Knowledge Base

This sample document is designed for testing a Retrieval-Augmented Generation (RAG) chatbot. It contains information about artificial intelligence, machine learning, databases, software engineering, and RAG systems so that you can ask questions and verify retrieval.

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computing concerned with creating systems that can perform tasks that normally require human intelligence. Common AI tasks include reasoning, learning, perception, language understanding, planning, and decision making.

Machine learning is a major area of AI in which algorithms learn patterns from data rather than being explicitly programmed with every rule. Deep learning is a subset of machine learning that uses multi-layer neural networks.

2. Machine Learning

Machine learning is commonly divided into supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labeled examples. Classification predicts categories such as fraud or not fraud, while regression predicts numerical values such as house prices. Unsupervised learning works with data without target labels and can be used for clustering or dimensionality reduction.

A typical machine learning workflow includes collecting data, cleaning data, exploring the data, selecting features, splitting the dataset, training a model, evaluating the model, and deploying the model.

3. Databases and SQL

A database is an organized collection of data. Relational databases store data in tables consisting of rows and columns. SQL, or Structured Query Language, is used to create, read, update, and delete data in relational database systems.

A primary key uniquely identifies a row in a table. A foreign key creates a relationship between tables. The SELECT statement retrieves data, WHERE filters rows, GROUP BY groups rows for aggregation, and ORDER BY sorts results.

SQL Concept	Purpose
SELECT	Retrieve data
WHERE	Filter rows
GROUP BY	Group rows for aggregation
ORDER BY	Sort query results
JOIN	Combine related tables

4. Software Engineering

Software engineering applies systematic methods to the development, testing, deployment, and maintenance of software. The Software Development Life Cycle (SDLC) commonly includes requirements analysis, design, implementation, testing, deployment, and maintenance.

Agile development emphasizes iterative delivery, customer feedback, collaboration, and adaptation to changing requirements. Scrum is an Agile framework that organizes work into short iterations called sprints and uses roles, events, and artifacts to coordinate development.

5. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, combines information retrieval with a generative language model. Instead of asking an LLM to answer only from its learned parameters, a RAG system first retrieves relevant information from an external knowledge base and then provides that information as context to the model.

A basic RAG pipeline contains document loading, text splitting, embedding generation, vector storage, similarity search, prompt construction, and response generation. Documents are divided into chunks because retrieving smaller passages usually makes it easier to provide focused context to the language model.

6. Embeddings and Vector Databases

An embedding represents text as a numerical vector. Texts with similar meanings can have embeddings that are close to each other in vector space. A vector database stores these vectors and supports similarity searches.

FAISS is a library for efficient similarity search over dense vectors. In a local RAG application, FAISS can store document embeddings and retrieve the chunks that are most similar to a user's question.

7. Example RAG Questions

- What is artificial intelligence?
- What are the three common types of machine learning?
- What does a primary key do?
- What is the purpose of GROUP BY in SQL?
- What are the major stages of the SDLC?
- What does RAG stand for?
- Why are documents split into chunks in a RAG system?
- What is an embedding?
- What is FAISS used for?

8. Key Terms

AI: Artificial Intelligence. **ML:** Machine Learning. **LLM:** Large Language Model. **RAG:** Retrieval-Augmented Generation. **SQL:** Structured Query Language. **FAISS:** Facebook AI Similarity Search.